

$$E = 30 \cdot 10^9 \text{ Pa}$$

$$\Delta l = 10^{-6} \text{ m}$$

$$\epsilon = \frac{\Delta l}{l}$$

$$\sigma = E \epsilon$$

$$\sigma = \frac{N}{A}$$

$$\frac{N}{A} = E \varepsilon \Rightarrow N = E A \varepsilon = E A \frac{\Delta l}{l}$$

$$N = 30 \cdot 10^9 \cdot \boxed{2} \cdot \frac{10^{-6}}{\boxed{2}} = \underline{3 \cdot 10^4 \text{ N}}$$

